

An Engineered Design of 3-axis Clinostat for Simulated Gravity Experiments in Astrobotany

Christian Rae C. Cacayurin^{1,*}, Juan Carlos L. De Chavez¹, Mariah Christa G. Lansangan¹, Chrischell B. Lucas¹, Justine Joseph D. Villanueva¹,
Ronnie Concepcion II^{1,2,3}, R-Jay Relano^{1,2}, Argel Bandala^{2,4}

¹Department of Manufacturing Engineering and Management, De La Salle University, Manila, Philippines

²Intelligent Systems Research Unit, Center for Engineering and Sustainability Development Research, De La Salle University, Manila, Philippines

³Plant and Soil Health Research Unit, Center for Natural Sciences and Environmental Research, De La Salle University, Manila, Philippines

⁴Department of Electronics and Computer Engineering, De La Salle University, Manila, Philippines

{christian_cacayurin, juan_carlos_dechavez, mariah_lansangan, chrischell_lucas, justine_villanueva, ronnie.concepcion, r-jay_relano, argel.bandala}@dlsu.edu.ph

Abstract—With the explorations of life in space, there is the need to sustain the basic needs of humans in space without constantly getting supplies from Earth. Hence, this is where space farming exudes its importance in today's time. Space farming, however, is not as similar to the farming practices on Earth due to the different environmental factors, especially gravity. There is a need for a device or facility that has the capability to mimic the space conditions on Earth. A clinostat is a device that comprises one or more axes that can rotate specimens and resemble partial gravity in a flight experiment. Hence, to study space farming, a clinostat is vital to be used in observing seedling tropisms. In the present time, this area of research could be considered sparse yet growing. With this, a conceptual design of a 3-axis clinostat is presented in this study with a dimension of 40 x 40 x 27 in. Moreover, this study suggests incorporating a rotary system, fertigation system, and lighting system in the conceptual design of the clinostat to better control the environment the seedlings will grow in. This study also presents the mechanical components of the proposed design and how the control system is integrated into the clinostat. The resulting clinostat may be used to cultivate a total of 36 seedlings which may be divided into two which may be subjected to two different light spectrums. The lighting system could help in observing the possible effects of different light settings on plants. Subsequently, the proposed fertigation system could help sustain plant growth by delivering water and nutrients to the roots through a spray mechanism.

Keywords—astrobotany, future food system, hypergravity, intelligent system, mechanical simulation, sustainable agriculture

I. INTRODUCTION

Agriculture forms the bedrock of global food production, and its sustainability is threatened by factors such as soil quality, water availability, extreme weather events, and climate change [1]. Space-based technology has emerged as a valuable tool for enhancing agricultural productivity and profitability [2]. Space agriculture, the practice of cultivating food and plants in space or on spacecraft like the International Space Station (ISS), is a growing field of study. While still in its developmental stages, it holds promise for both space missions and Earth's food supply chain. Researchers are working on making space greenhouses large enough to support long-term human needs. This

innovation in space agriculture has the potential to significantly impact food production, benefiting both space exploration and Earth's agriculture.

To advance the understanding of space agriculture, various devices like clinostats are used to simulate the effects of gravity in space [3]. Clinostats, which rotate plants or biological samples around an axis perpendicular to gravity, help researchers study plant responses to microgravity conditions. The development of equipment that accurately replicates weightlessness is crucial for improving the understanding of plant growth in space [4]. These tools enable scientists to explore the physiological and developmental aspects of crops, particularly how they adapt to the absence of gravity and interact with their environment.

For astronauts on space stations, a sustainable food supply is currently dependent on regular shipments from Earth [5]. However, as missions venture farther from our planet, the need for self-sustaining food production becomes increasingly vital [6]. Space farming is poised to play a pivotal role in addressing this challenge. Understanding the physiology and significance of crops is critical for successful space agriculture.

This study involves designing a 3-axis clinostat for simulating gravity in experiments. This specialized device is able to replicate space-like gravity conditions in a controlled setting. With its three axes, complex gravitational forces can be mimicked to understand how living things react to altered gravity. The research aims to improve the area of research regarding the devices that could be utilized to further understand the implications of gravity conditions outside Earth. For this study, the primary objective is to present a conceptual design of a clinostat incorporated with a rotary system, fertigation system, and lighting system that could be used in gravity simulation experiments.

II. DESIGN CONSIDERATIONS BASED ON SCIENTIFIC LITERATURES

A. Clinostat as Gravity Simulating Device

A clinostat is considered to be one of the most effective tools for imitating microgravity conditions. Experiments for the

simulation of gravity, such as partial gravity and microgravity, on Earth are mostly done by using clinostats and random positioning machines (RPM). Concomitantly, these devices are capable of simulating the value of microgravity and partial gravity needed in a particular experiment. However, one notable characteristic of clinostat is its capability to produce constant rotation as compared to other gravity-simulating devices. Clinostat is a machine that constantly rotates a sample to imitate microgravity, thus achieving theoretical weightlessness once a rotation is completed [7].

This study focuses on using clinostat as a gravity-simulating device. To expound, in [8], removing the effects of gravity was achieved by utilizing a clinostat where samples are rotated at constant speed in two different axes. Experiments such as [9] have also used a clinostat to test the viability of integrating microgravity and aeroponic technology on Earth. Moreover, [9] focuses on using a three-axis clinostat in observing and supporting plant growth where the concepts of quasi-weightlessness and aeroponic process are combined. Subsequently, many studies have observed the possible effects of microgravity, partial gravity, and hypergravity in plants. One of which is a study by [10] that discusses the damage induced to antioxidant metabolism and oxidative in *Anthemis gilanica* cells when exposed to fast clinorotation. In this study, the clinostat utilized was two-dimensional. Furthermore, to test the samples, the rotation was set to 30 rpm for seven days. It was observed in [10] that the eradication of oxidative damage on macromolecules is feasible when subjected to clinorotation together with the induction of enzyme activity in the phenylpropanoid biosynthesis pathway and antioxidant capacity. Moreover, the cell growth, viability, protein content, DPPH, and antioxidant power also increased in [10] due to clinorotation. These studies show that utilizing a clinostat as a gravity-simulating device is effective.

To further emphasize the clinostat's impact on plants, a study by [11] demonstrates how it could affect plant growth. A 2D clinostat was used in this study, and the results showed that the absence of gravity caused by the clinostat prevented a plant from growing with a hook-like structure. Hence, it is beneficial to further observe the effects of the absence of gravity on plant growth. Similarly, in studying the physiological and biochemical changes in cherry tomatoes when introduced to microgravity, [7] created a clinostat with only one mechanism with three degrees of freedom. This study utilized only one motor to produce an economical and accessible clinostat. Ergo, a clinostat's design could be easily modified according to its use.

B. Parameters in Gravity Simulation Experiments

To design a clinostat effectively, different parameters should be considered. Table 1 shows the specific dimensions, range of g value, and range of rotation in a specific study with the goal of simulating either partial gravity or microgravity by utilizing clinostat. Utilizing these parameters ensured that the clinostat would be able to meet each study's objectives. Overall, most studies utilized a speed value of 1 rpm to 10 rpm. It is shown in the table below that the maximum value used is 120 rpm. Moreover, the g value gathered ranges from 0g to 1g, while the dimensions differ according to the design of the clinostat and the main objective of the study.

TABLE I. RELATED GRAVITY SIMULATION EXPERIMENTS

Dimension of Simulating Platform	Range of Gravity Value	Range of Motor Rotation	Reference
No Specified	$\approx 0g$ to $1g$	1 rpm to -1 rpm (reverse direction)	[12]
20 cm radius	$6 \times 10^{-3}g$ to $17.8 \times 10^{-3}g$	20 cm radius	[13]
Chamber's diameter is 30 cm	Not Specified	120 rpm	[14]
Petri Dish's diameter: 9 cm	0.00005g to 0.18g	1 rpm (slow clinorotation) to 60 rpm (fast clinorotation)	[15]
Acrylic cylindrical housing: 9.4cm diameter x 9.0 cm height	$8.9 \times 10^{-3}g$	20 rpm	[16]
Seedling Tray: 41.3 cm x 14.5 cm	Not Specified	2 rpm to 10 rpm	[17]

C. Space Farming Technologies

Space exploration and missions are actively growing areas of research and life sustaining research, and technologies are also on the rise to support other space missions. Space farming is an area of research that is concerned with growing plants or crops in environments outside of Earth. The objective of space farming research is to be able to efficiently and continuously produce food for humans in space or in other celestial bodies such as Mars and the Moon. In doing so, it is important to find the optimal environment possible for any crop to grow on. This implies an extensive consideration and study of the effect of different cultivation methods and environmental factors. Shown in Table 2 is an overview of how there are different ways to cultivate crops. Different farming strategies may be applied to support plant growth as long as plants are supplied with the nutrients, they need in order to grow. Shown in Table 3 on the other hand, are some of the systems designed by studies to stimulate and maintain plant growth.

TABLE II. PLANT CULTIVATION METHODS

Crop	Growth Media	Cultivation Setup	Reference
Chinese Cabbage	Calcined Clay	Two seeds of Chinese cabbage were set in a drum container of calcined clay mixed with fertilizer in a ground-based veggie-analog chamber. Seeds were subjected to 28 days of cultivation and there is an observed decrease in terms of plant productivity when exposed to increased levels of CO_2 similar to the ISS.	[19]
Corn Wheat Pea Rice	Agar Medium	A one-axis clinostat was used with a rotational axis angle of 90 degrees, rotating at 90 rpm. Seeds of pea, rice, corn, and wheat were cultivated in agar medium for 5, 6, 9, and 5 hours respectively. Enhancement in amino acid concentration is observed in all plants compared to control.	[20]

Crop	Growth Media	Cultivation Setup	Reference
Cucumber	Water-absorbable plastic foam	Seedlings were placed in pockets made of plastic foam and were enclosed in a plastic container together with a distilled water chamber and NaCl chamber. Seeds were left to germinate for 18 hours and then were subjected to 3D clinorotation at an angle of 90° in a dark environment with $25 \pm 1^\circ\text{C}$ temperature. The seedling roots showed a positive hydrotropic reaction in the absence of gravitropism.	[21]

TABLE III. SYSTEMS FOR SUSTAINING PLANT GROWTH

System Name	System Set-up Description	Reference
Nutrient root-feeding system	An advanced nutrient-feeding system is designed for plant growth facilities for the ISS. The study is mainly concerned with feeding nutrients in the roots by releasing them in fertilizer. A controller regulates the solution flow and levels of nitrogen, potassium, phosphorus, and granular mineral-rich ionite.	[22]
Environment control system	A control system is designed to monitor the temperature, humidity, light, and gas levels inside the chamber of a 3-axis clinostat. The chamber is also equipped with a data control display. This control system is controlled by three motors run by an Arduino microcontroller.	[23]
Light system	This study compared the growth of slime molds in petri dishes containing non-nutrient agar subjected to different light wavelengths. Forty plates were prepared and grouped into four. Each set was wrapped in a black party streamer and placed in a clinostat. The chamber of the clinostat has three specific light spectra – red, green, and blue. The plates were tested for red, blue, green, red/blue, and dark.	[24]

III. CONCEPTUAL DESIGN OF 3-AXIS CLINOSTAT FOR GRAVITY SIMULATION EXPERIMENTS

This section covers the techniques and processes that must be followed in sequential order to create a device capable of imitating gravity conditions outside Earth. The architecture presented in Fig. 1 focuses on the design aspect of the clinostat.

A. Rapid Design and Prototyping of Clinostat

A 3D model of the chassis and clinostat is designed and shown in Fig. 2. The clinostat has three degrees of freedom and has a total dimension of 40 inches in length and width and 27 inches in height. The material set for the chassis is acrylic while the material for the clinostat is aluminum.

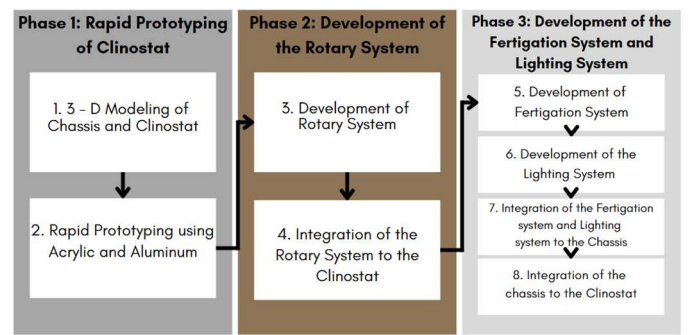


Fig. 1. Methodological Architecture for the 3-axis Clinostat for Partial and Light Interactions in Seedling Tropisms

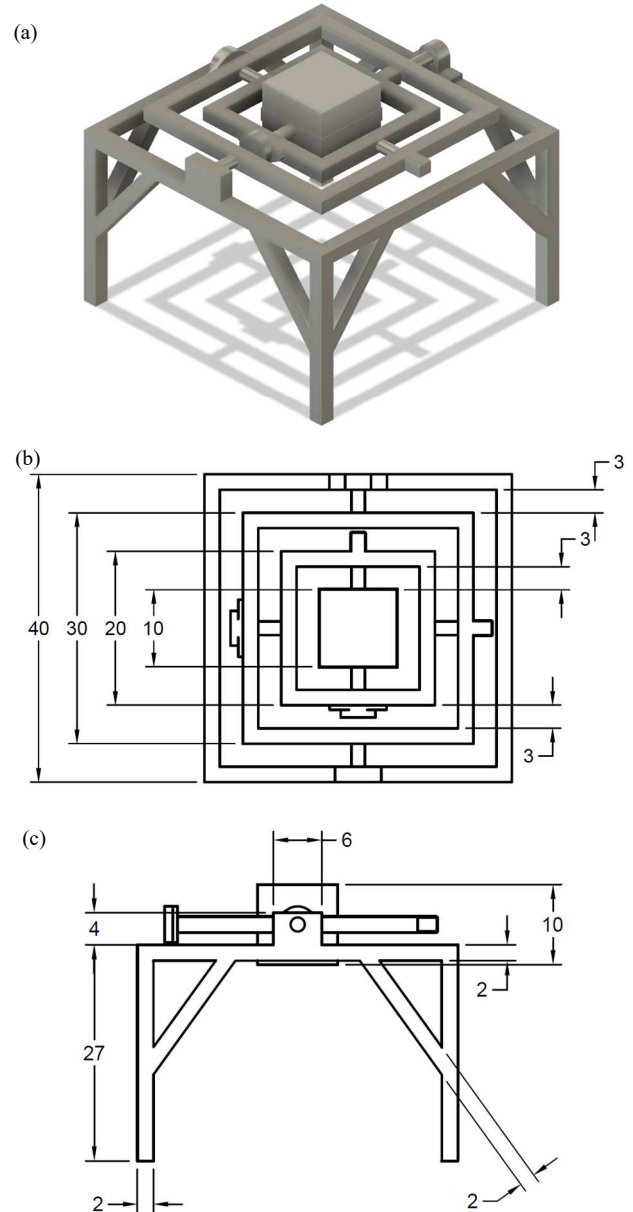


Fig. 2. (a) Conceptual design of clinostat and the dimensions showing the (b) top and (c) side views.

B. Development and Integration of Rotary Control System

The development and integration of the rotary system of the clinostat is a crucial part of the process since it involves the system which is responsible for the simulation of partial gravity. The rotation of each axis is illustrated in Fig. 3(a), while the placement of the corner bevel gear, bevel gear, pillow block bearing, flange bearing, and slip ring commutator can be seen in Fig. 3(b). The rotary system suggested includes wiper motors that could be responsible for the actuation of the 3-axis. This could be controlled by a PWM (Pulse Width Modulation) and could be powered by a 12 V and 12 A power supply. To avoid friction as much as possible, bearings should be utilized.

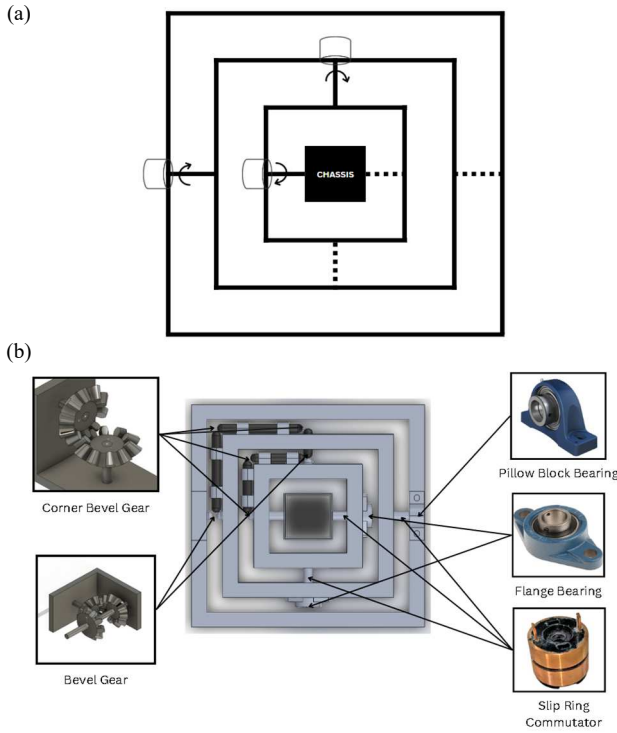


Fig. 3. (a) 3-axis rotation nomenclature highlighting three rotors with the cultivation chamber (chassis) at the center of the rotating platform. (b) Pictorial structure of mechanical elements for continuous rotary motion.

C. Development of the Fertigation System and Lighting System

The development and integration of both the fertigation system and lighting system are responsible for the automated fertigation of the plants inside of the clinostat, as well as the lighting system to make sure that the plants will get the light that they need, for the fertigation and lighting system (Fig. 4(a)). It could be composed of a water reservoir that could hold about 500 ml of water, watering pumps that could supply adequate amounts of water to roots, moisture sensors that could track the moisture levels which will trigger the water pumps to spray whenever there is a lack of moisture (Fig. 4(b)). The lighting that could be utilized is an LED strip, which is capable of adjusting the color of the light needed accordingly (Fig. 4(c)). The automated lighting system will consist of different artificial light setups, mainly pure red, pure blue, red and blue, and white LED strip colors.

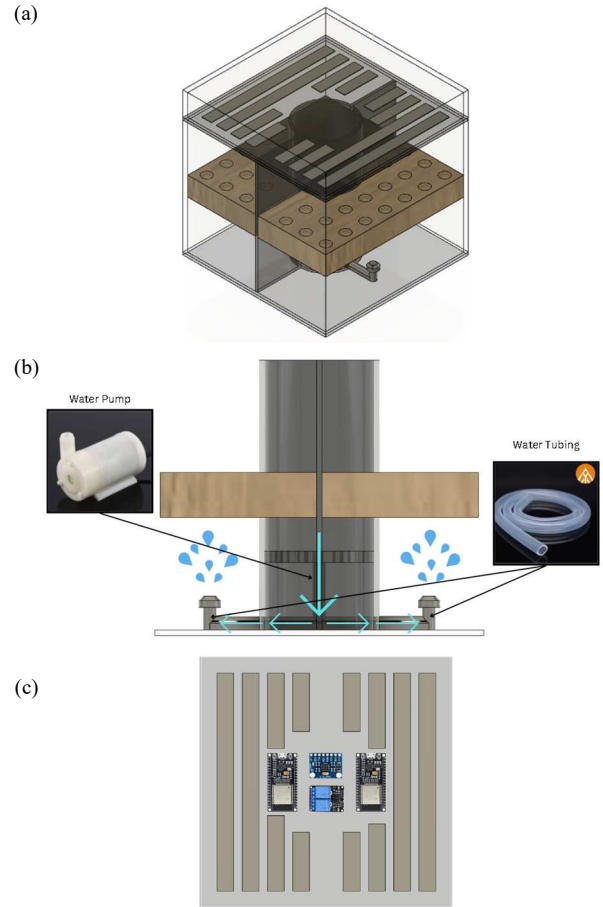


Fig. 4. (a) Integration of the fertigation system and lighting system to the chassis. (b) Flow of the fertigation system consisting of reservoir, plumbing, and sprinkler. (c) Placement of microcontrollers, sensors and relay driver for artificial lighting system and fertigation system.

IV. DISCUSSION

The design of this study's clinostat was based on the aerotrim which is a 3-axis gimbal that is used by astronauts for their anti-gravity training, it has three different axes rotating in different directions with a specific rpm which will simulate hypergravity, the rotation of each axis is controlled by a single motor, where the rotation of the motor is transferred to the three axes by gear trains. Concomitantly, each axis needs a specific rpm to simulate hypergravity. With that said, the gear trains mentioned adjust the rpm according to the specification needed for each axis, which means that the rpm of each axis is only limited to the rpm set by the gear train and cannot be set individually. The researchers considered to have individual motor per axis which will allow the axis to have a variable rpm that can be set according to the desire of the user. However, the problem that the researchers have encountered is that since the clinostat rotate the wiring of the individual motor, there will be difficulties even if a slip ring is utilized. This problem could persist due to the reason that the slip ring only has a limited number of wires, and the said wires are used for the automated fertigation system inside of the chamber of the clinostat.

The difficulty of maintaining stability throughout the experiment is also one significant limitation in the present design. Problems arising from mechanical vibrations or

deviations from an accurate rotating axis may affect the clinostat. Improvements in materials and manufacturing approaches, such as the use of durable yet lightweight components and precision machining, might help reduce mechanical vibrations and improve overall stability to overcome this constraint and improve the stability of clinostats. Through integrating it, researchers may improve the precision and dependability of clinostats, assuring greater durability and repeated experiments in situations that simulate hypergravity.

On the other hand, the fertigation system of the clinostat was designed to automatically fertigate the plants inside of the chamber. Since the chamber will be rotating constantly for two weeks, the researchers have decided to use a horticultural foam. This will reduce the weight of the chamber and be easier to manage than soil. Additionally, a LED strip is also utilized to simulate light. The LED strip use is a variable type so that the researchers can also assess whether there is a difference of growth when expose to different colors of light. The automated fertigation system is mainly controlled by an Arduino ESP32 since this Arduino has built in Wi-Fi, monitoring the state of the chamber through a moisture sensor is possible, which is also responsible for starting the process of the automated fertigation system. Additionally, to power the electronics inside the chamber, a slip ring is used in to have a consistent power supply. To further improve the design of the fertigation system of this clinostat, the researches suggest that additional sensor should be placed inside of the chamber to gather as much information as possible from the experiment. Another suggestion is to include an additional drainage system in case that the plant doesn't absorb the water fast enough.

Subsequently, this conceptual design utilizes three axes and integrates automation in executing the environmental factors needed to replicate environmental conditions in space. Due to its flexibility and ease of use, this could be deemed advantageous as compared to other designs of clinostats. Consequently, there is an opportunity to delve deeper into the study of spaceflight experiments/astrobotany experiments here on Earth. This implies that we can simulate or replicate certain aspects of space conditions in controlled environments on our planet. By doing so, we aim to enhance our understanding of how the unique environments of other planets might impact living organisms. The main goal is to investigate and comprehend the potential effects of these extraterrestrial environments on living things. This could involve studying the physiological, biological, and ecological responses of organisms when exposed to conditions similar to those found on other planets. Such experiments would contribute to our knowledge of the challenges and opportunities associated with space travel. This research could go beyond mere observation and analysis. It hints at a proactive approach by proposing to explore how we can apply the insights gained from these experiments. Thus, contributing to the limited studies available regarding spaceflight experiments and astrobotany experiments. The focus is on identifying environmental factors that could be beneficial for living organisms. This implies the possibility of using the knowledge gained to create conditions that support and promote the well-being of living things, whether on Earth or potentially on other planets in the future.

In the perspective of applied and experimental astrobotany, it would be more ideal to utilize a platform that could simulate

both artificial microgravity and hypergravity [25] as the space shuttle experiences hypergravity when escaping the Earth's gravity and an immediate turn to microgravity when it is already out of Earth's pull. The use of intelligent optimization algorithms and advanced statistical metaheuristics [26] in determining the best gravity value that could stimulate the growth of plants even when stressed with other abiotic and biotic stimuli. Because the roots of plants are too sensitive to drying and its lateral strands might be punctured or bend when pulled out from its growing medium for validation of its growth, it is therefore important to utilize non-destructive digital phenotyping techniques and technologies such as bioimpedance and capacitive resistivity [27-29]. This increases the integrity of real root growth as the chances of damaging and altering its natural development will be avoided. In the standpoint of energy requirement, the use of bioenergy such as coming from microalgae [30] could potentially support the mid-to-high power requirement of a rotating platform. On top of this are the irrigation and fertigation systems [31] which should be intelligent in providing just enough liquids to the growing crops and still comply to green economy and circular economy in real space-like situation. Overall, a simulated gravity platform for astrobotany applications is a complex system and complex environment but designing it based on engineering concepts would not only commit to the biological requirements of cultivating multiple crops but also advances the understanding of how gravity could really support fresh food production when humans have successfully colonized in space.

V. CONCLUSION AND RECOMMENDATION

The development of a 3-axis soilless clinostat can result in remarkable findings and advancements that show significant opportunities for further research within astrobotany and future occupation in space. The integration of essential components and configurations, such as the fertigation system, lighting system, and rotary system, serve a significant role in the achievement of the study and also establish a solid basis for future studies. The fertigation system that has been developed to specifically provide accurate and controlled nutritional solutions to seedlings. This system serves as a valuable tool for researchers, enabling them to investigate the complicated interactions between nutrient availability and seedling tropisms under settings of partial gravity. In addition, the implementation of the advanced lighting system provides with the capability to exert exact control over various light factors. This allows them to conduct detailed examinations into the influence of light on the development of plants in situations with lower gravity. The clinostat serves as an important component in our experimental configuration, offering a very beneficial framework for the simulation of partial gravity environments. The versatility and precision of this technology enable researchers to investigate various degrees of gravity and their impact on seedling tropisms, therefore enhancing our comprehension of plant responses to modified gravitational conditions. Different parameters used in the studies mentioned above are rendered beneficial in observing the possible impact of gravity simulation experiments on living organisms. Through these, it was observed that both 2D and 3D clinostats are efficient in imitating gravity conditions outside Earth. Consequently, the effect of the absence of gravity varies per plant. This variation roots from the differences in

setups and nutrients injected into a specific crop or seedling. For future works, it is recommended to apply life cycle assessment in analyzing the environmental emissions caused by fabricating such simulated gravity platform.

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